Abhishek Katarapu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	5.76
Sri Chaitanya Co education College Gayathrinagar	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	88.50%
Oxford EM High school	Class X (Board of Secondary Education, Andhra Pradesh)	2021	95.00%

EXPERIENCE

Freelance Engineering Developer | TVS Motor Company |Jun 2026 - Jul 2026

- Analysed system-level requirements for TVS iQube EV digital-twin development, contributing to **8+ engineering deliverables**
- Executed testing, integration and validation cycles across **4+ subsystems**, identifying and resolving **12+ defects**
- Documented validation results and technical specifications across **7+ engineering reports**, supporting EV development and verification workflows

Founder & Full-Stack Developer | UpcycleX | Jan 2025 - Sep 2025

- Launched an AI-enabled thrift-fashion marketplace using **React, Supabase and PostgreSQL**, supporting **100+ users and 300+ product listings**
- Designed and implemented **15+ REST APIs** with JWT authentication for user accounts, product listings, profiles and marketplace operations
- Modelled a **10+ table relational database schema** and owned the product lifecycle from architecture and development through deployment

SKILLS

Design & Analysis: CAD, SolidWorks, CATIA, Finite Element Analysis (FEA), ANSYS, MATLAB, Simulink, Product Design, Mechanical Design, GD&T, Design for Manufacturing (DFM)

Core Engineering: Control Systems, Fluid Mechanics, Thermodynamics, Heat Transfer, Machine Design, Manufacturing Processes, Computational Fluid Dynamics (CFD), EV Systems, Automotive Systems

Testing & Validation: Testing & Validation, Root Cause Analysis, Technical Documentation, System-Level Testing

Prototyping & Embedded: Arduino, Motor Control, Sensor Integration, Feedback Control, Embedded Systems, Rapid Prototyping

PROJECTS

2D Robotic Arm Control System | Arduino, Sensors, Actuators, Feedback Control

- Built **2D robotic-arm prototypes** integrating **sensors, actuators, and Arduino control** for controlled positioning and coordinated motion
- Implemented **closed-loop position feedback** and control logic for **trajectory tracking and automated motion control**
- Evaluated **positioning accuracy, repeatability, and response time**, achieving **±2 mm positioning accuracy**

Rod Surface Smoothener | Mechanical Design & Embedded Control | Arduino, Motor Control

- Designed a **rod surface-finishing prototype** integrating a mechanical finishing mechanism with **programmable motor speed control**

- Programmed **Arduino-based control logic** to maintain consistent finishing speed under varying operating conditions
- Tested the prototype across **10+ operating conditions**, achieving **~25% improvement in surface finish consistency**

NexusCare | AI Healthcare Triage Platform | Python, ML, NLP, FastAPI, PostgreSQL

- Trained an **ML/NLP triage model** to classify symptom cases into **3 urgency tiers**: routine, priority, and emergency
- Developed a **FastAPI + PostgreSQL backend** with **10+ REST APIs** for symptom history, case summaries, and appointment management
- Validated the model at **91% test accuracy** across **1,000+ held-out cases**, evaluating performance across all **3 urgency tiers**

EXTRACURRICULAR

- Developed interdisciplinary projects combining **Mechanical Engineering, Control Systems, Software Development and Machine Learning**.
- Built independent software and engineering prototypes involving **automation, APIs, databases and embedded systems**.